

Red Hat Lightspeed 1-latest

Using APIs to configure Red Hat Lightspeed services

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Abstract

This document describes the web services available to integrate Red Hat Lightspeed with external applications. The Red Hat Lightspeed RESTful APIs are implemented as standard REST HTTP requests and responses of content type JSON.

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CHAPTER 1. INTERACT WITH RED HAT LIGHTSPEED APIS

Red Hat Lightspeed is a predictive analytics service that helps identify and remediate security, performance, and availability threats to your Red Hat infrastructure. Red Hat Lightspeed is built on APIs (Application Program Interfaces) that supply data and functionality to its various applications.

The Red Hat Lightspeed front end (the user interface, or UI) directly communicates with the back end API (the server) to retrieve and display information. You can interact with Red Hat Lightspeed through the UI, or create your own custom clients to connect directly to the back-end API.

You can use API commands to interact with web services to perform various operations, such as retrieving data, sending data, updating records, and deleting records. Using APIs to interact with Red Hat Lightspeed enables you to perform Red Hat Lightspeed operations synchronously or asynchronously, depending on the service and endpoint. The Inventory export service is an example of an asynchronous operation, but most operations work synchronously. You can incorporate API calls into your web-based applications, and you can even automate them.

Using APIs presents many advantages, including:

- Improved security – You can create a structured URL that the user can use to request information. This helps to protect sensitive data on a server.
- APIs allow computers to interact directly with other computers.
- APIs can be scripted. You can configure automation to run scripts on-demand.

Red Hat Lightspeed APIs follow the OpenAPI specification. By default, all Red Hat Lightspeed API resources are represented in **application/json** format. You can also export reports, inventory, and maintenance plans as **text/csv**.

1.1. ABOUT REST API AND HTTP METHODS

Representational State Transfer (REST) API is a set of rules and conventions for building and interacting with web services. REST APIs use URLs as resources to communicate with servers by sending HTTP requests.

HTTP methods perform CRUD (Create, Read, Update, and Delete) operations on API resources. The most common HTTP methods used are:

- GET (retrieve data)
- POST (send data)
- PUT (insert data)
- DELETE (remove data)
- PATCH (update data)

An API request (also known as an API call) requests that the server perform the operation that the HTTP method describes (for example, GET) from the location that the request specifies.

Additional resources

- OpenAPI specification

1.2. RESPONSE CODES

APIs return numerical status codes that indicate whether an API call was successfully processed by the server. You can identify why the call fails, using a status code from a server that describes the reason for the failure.

The following table shows types of API response codes:

Response code	Response	Description
100/1xx	Request received	Informational codes. The server has received the request.
200/2xx	Success	200 = The server has successfully processed the request. 2xx = The request has succeeded, and more information is available.
300/3xx	Redirect	The URL/URI of the requested resource exists at a different location than the one you requested.
400/4xx	Client Error	The request failed. 400 = Bad request 404 = Server not found/resource not found 4xx = The request failed, and the cause of the failure is described.
500/5xx	Server Error	The server cannot handle the request.

Additional resources

- HTTP status codes
- OpenAPI specification

CHAPTER 2. CONFIGURE AUTHENTICATION WITH RED HAT LIGHTSPEED APIS

You can authenticate to Red Hat Lightspeed APIs by using token-based service accounts or certificate-based authentication for API call automation.

- Token-based service accounts
- Certificate-based authentication

2.1. CONFIGURE SERVICE ACCOUNT ACCESS

You can create and configure a service account, then generate a client ID and client secret pair to use with that service account.

Procedure

1. From the Red Hat Hybrid Cloud Console, click the **Settings** icon and select **Service Accounts**.
2. Click **Create service account** to set up the account.
3. Enter a service account name and a short description, and click **Create**. The Hybrid Cloud Console creates the service account and displays the unique client ID and client secret for the account.
4. Copy the generated client ID and client secret values to a secure location. You need to specify these credentials when configuring a connection to a service.

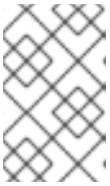


IMPORTANT

The client secret displays *only once*, so ensure that you have successfully and securely saved the copied credentials before closing the credentials window.

5. After you save the client ID and secret to a secure location, select the confirmation checkbox in the credentials window and close the window. The service account and its client ID appear on the Service Accounts page.
6. Add the service account to the applicable User Access groups. NOTE: Service accounts do not inherit permissions from the Default access group, so you must place them in a User Group to grant them access permissions.
7. At the command line, use a **curl** command with Red Hat single sign-on (SSO) to generate an access token. Use the service account credentials that you generated in place of **<client_id>** and **<client_secret>**.

```
curl https://sso.redhat.com/auth/realms/redhat-external/protocol/openid-connect/token -d "grant_type=client_credentials" -d "scope=api.console" -d client_id=<client_id> -d client_secret=<client_secret>
```



NOTE

For more information about Red Hat SSO and generating access tokens, see Transition of Red Hat Hybrid Cloud Console APIs from basic authentication to token-based authentication via service accounts (Red Hat Knowledgebase).

8. Red Hat Single Sign-On (SSO) responds with a JSON object containing an access token. You can use this access token to make authenticated requests to APIs. To make a request, include the access token generated in the previous step in an Authorization: Bearer HTTP header in your request. Substitute the access token for **<access_token>**.

```
Authorization: Bearer <access_token>
```

Additional resources

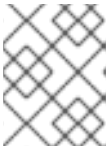
- Managing group access with roles and members
- Transition of Red Hat Hybrid Cloud Console APIs from basic authentication to token-based authentication via service accounts (Red Hat Knowledgebase)

2.2. GENERATE AN OFFLINE ACCESS TOKEN

The API queries require an active access token. To generate the access token, complete the following steps.

Procedure

1. Log in to the Red Hat Customer Portal with your username and password.



NOTE

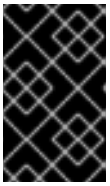
You can no longer log in to your Red Hat account with your email unless your email is your Red Hat login. For more information, see Finding your Red Hat login .

2. Navigate to the Red Hat API Tokens page .
3. Click **Generate Token** to generate an offline token.
4. At the command line, use a **curl** command to generate an access token. Use the offline token that you generated in place of **<offline_token>**.

```
curl https://sso.redhat.com/auth/realms/redhat-external/protocol/openid-connect/token -d
grant_type=refresh_token -d client_id=rhsm-api -d refresh_token=<offline_token>
```

5. Copy the generated access token and save it in a secure location. You can use this access token to make authenticated requests to APIs.
6. To make a request, include the access token in an **Authorization: Bearer** HTTP header in your API call. Substitute the access token for **<access_token>**.

```
Authorization: Bearer <access_token>
```



IMPORTANT

Save the token. The Red Hat API Tokens page displays each token *only once*, and does not store tokens. Offline access tokens generated from the Red Hat API Tokens page expire after 30 days of inactivity.

Additional resources

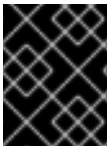
- Transition of Red Hat Hybrid Cloud Console APIs from basic authentication to token-based authentication via service accounts (Red Hat Knowledgebase)
- Creating and managing service accounts

2.3. ADDITIONAL RESOURCES

- API Cheat sheet

CHAPTER 3. MAKE API CALLS

You can make API calls to Red Hat Lightspeed APIs to automate tasks and access service data. Use **curl** commands, the Red Hat Developers API Catalog, the legacy API documentation, or programming language interpreter in your client code.



IMPORTANT

You must set up and configure a service account for authentication in the application or system that will perform the API calls.

The Red Hat Developers API Catalog helps you to create valid API calls in a variety of languages and formats, including **curl** commands. For example, you can send an API call in a **curl** command or embed a Python-formatted API call in Python code. Using the **GET/api/insights/v1/account** API call in the command or in code returns a list of all known accounts by Organization ID **org_id**.

3.1. USE THE CURL COMMAND FOR API REQUESTS

You can use the **curl** command to communicate with a server over HTTP without having to use a browser. **curl** is useful for making quick requests from the command line, such as requesting status, making updates to parameters, or requesting the values for specific parameters.



NOTE

This example procedure uses the Advisor API endpoint.

Prerequisites

- You have an access token that you obtained from your service account or from an offline access token for authentication.
- The service account has the necessary permissions to interact with the requested endpoint.

Procedure

1. Open the API catalog in a web browser. The API catalog lists available APIs for Red Hat Lightspeed and other services.
2. Select **Advisor** from the catalog. The Red Hat Lightspeed Advisor API page opens. The page lists the base URL of the Advisor API and any requirements for authentication.
3. Under the Operations list, click the drop-down arrow next to **GET /api/insights/v1/account**. The description for the operation includes parameters you can use to refine your API call, and also expected responses from the server. In addition, the panel on the right side of the page has sample code to perform the API call in many languages.
4. In the panel, click the drop-down and select **CURL** from the list of options. The panel displays sample code for the GET operation, formatted as a **curl** command.
5. Copy the sample code and paste it into your command line or into an existing code file where you want to start the GET command.

```
# curl --request GET \
```

```
--url https://console.redhat.com/api/insights/v1/account/ \
--header 'Authorization: Bearer <token>' \
--header 'Content-Type: application/json'
```

6. Paste your access token in place of **<token>**.

7. Press **Enter** to start the command.
   If the command line returns a response of 200, your API call was successful. In addition to a successful response and an optional description of the operation, the server returns the data you requested.

   If the command line returns a response other than 200, refer to Response Codes to determine how the API call failed and how to remedy the reason for the failure.

Additional resources

- Red Hat Lightspeed API cheat sheet
- Red Hat Developers API catalog

3.2. USE THE RED HAT LIGHTSPEED API CATALOG AND DOCUMENTATION

The Red Hat Lightspeed API Catalog and Documentation page in the Red Hat Hybrid Cloud Console describes each operation you can perform with the Red Hat Lightspeed API. You can create API requests in multiple languages and formats, copy the requests, and then use them within your own code to make the API calls.

The Red Hat Developers API Catalog shows you how to create valid API calls in a variety of languages and formats, including python code. For example, the **GET/api/insights/v1/account** API call returns a list of all known accounts by Organization ID **org_id**.



NOTE

This example procedure uses the Advisor API endpoint.

Prerequisites

- You have the programming language you want to use (for example, python) installed on your system.
- You have an access token that you obtained from your service account or from an offline access token.
- You have a service account configured with necessary permissions.

Procedure

1. Open the API catalog in a web browser. The API catalog lists available APIs for Red Hat Lightspeed and other services.
2. Select **Advisor** from the catalog. The Red Hat Lightspeed Advisor API page opens. The page lists the base URL of the Advisor API and any requirements for authentication.

3. Under the **Operations** list, click the drop-down arrow next to **GET /api/insights/v1/account**. The description for the operation includes parameters you can use to refine your API call, and expected responses from the server. In addition, the panel on the right side of the page generates an example API call for the GET operation in multiple programming languages.

4. In the panel, click the drop-down and select the programming language that you prefer from the list of options. The panel displays sample code for the GET operation, formatted in Python syntax.

5. Copy the sample code and paste it into a Python code file where you want to call the GET command. For example:

```
import requests

url = "https://console.redhat.com/api/insights/v1/account/"

headers = {
  "Authorization": "Bearer <token>",
  "Content-Type": "application/json"
}

response = requests.get(url, headers=headers)

print(response.json())
```

6. Paste your authorization key for authentication in place of **<token>**.

7. Press **Enter** to run your code.
   If the server returns a response of 200, your API call was successful. In addition to a successful response and an optional description of the operation, the server returns the data you requested.

   If the code returns a response other than 200, refer to Response Codes to determine how the API call failed and how to remedy the reason for the failure.

Additional resources

- Red Hat Lightspeed API Catalog and Documentation
- Red Hat Developers API Catalog

3.2.1. View data formats and schemas

The data formats and schemas for Red Hat Lightspeed API calls display the formats and types for data returned with server responses. You can use this information in your code and for troubleshooting unexpected results.

The following example uses the advisor service.

Procedure

1. Open the API catalog in a web browser. The API catalog lists available APIs for Red Hat Lightspeed and other services.

2. Select **Advisor** from the catalog. The Red Hat Lightspeed Advisor API page opens. The page lists the base URL of the Advisor API and any requirements for authentication.

3. Under the Operations list, click the drop-down arrow next to **GET /api/insights/v1/account**. The Operations section contains the query parameters you need to specify in an API call and the expected format of each parameter (for example, **integer**).

4. To see the format for the server response, click the drop-down arrow next to the **Responses** section. The section shows an example of data that the API returns with a successful response.

5. To see the formats for the returned data, click the **Schema** link next to the server response (for example, **PaginatedOrgIdList**). The page shows the schema for the data in the response, and the formats and types of all of the components that make up the data (for example, **org_id** is a string, and **count** is an integer).

3.3. USE THE LEGACY API DOCUMENTATION

The Red Hat Hybrid Cloud Console contains autogenerated legacy Swagger documentation for Red Hat Lightspeed API operations. For your automation purposes, you can customize the provided example code for API calls, execute the requests from within the documentation, and receive the server response code.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.
- You have the necessary permissions to perform the queries.

Procedure

1. Navigate to the API documentation page in the Red Hat Hybrid Cloud Console.

2. Select the application that you want to use from the list (for example, **Advisor**). The page for the application displays, with a list of all the operations you can request for that application.

3. Select the operation that you want to perform. For example, navigate to **Advisor → account → GET /api/insights/v1/account**.

4. Click the drop-down arrow for the operation. The Swagger documentation page describes the operation and lists the parameters you can use to refine your request. The page also shows the JSON format for a successful server response.

5. Click **Try It Out**. The page displays the format of a successful server response. If the operation you selected requires a value for a parameter (for example, **org_id**), enter the value in the blank.

6. Click **Execute**. The page displays a **curl** request to the server, the request URL, and the server response.

    a. To copy the **curl** request to the clipboard, click the clipboard icon on the right side of the **curl** request.

    b. To copy the server response to the clipboard, click the clipboard icon in the top right corner.

    c. To download the server response as a JSON file, click **Download**.

7. Optional: To clear the server response and retry the command, click **Clear**.

Next steps

- You can use the generated requests in your own code to perform API calls to the Red Hat Lightspeed API server. For POST requests, you can enter formatted data into the **Request Body** section of the request and select the format for the data, for example, **application/json**.
- For detailed information about the parameters, schemas, and descriptions for each operation, click the **View JSON** or **RAW** links on the **API Documentation** page. Both links display the operations, required formats for parameters, detailed descriptions for each operation, and the required information in each schema. The **RAW** link also displays information about the header for both server requests and responses.

Additional resources

- Red Hat Lightspeed API cheat sheet

3.4. ADDITIONAL RESOURCES

- Red Hat Developers API Catalog
- Chapter 2, *Configure authentication with Red Hat Lightspeed APIs*